

CreditPathAI - Automating and optimizing the loan recovery lifecycle by modeling repayment behavior using diverse data.

Project Objective

To design and develop a machine learning–driven platform that predicts borrower default risk and recommends personalized recovery actions.

using open-source technologies to ensure cost-effectiveness, scalability, and reliability.

The system aims to improve delinquency recovery efficiency and equip collection agents with actionable insights.

Project Workflow

1. Data Ingestion

- Load datasets from Kaggle Loan Default Dataset and Microsoft R Server Loan Credit Risk.
- Tools: CSV/Excel ingestion, SQLite/PostgreSQL.

2. Feature Engineering

- Perform cleaning, normalization, joining, and feature extraction (e.g., repayment velocity, credit utilization).
- Tools: Pandas, Dask (for large datasets).

3. Model Training & Evaluation

- Algorithms: Logistic Regression (baseline), XGBoost, LightGBM.
- Metrics: AUC-ROC, Confusion Matrix.
- Tools: Python, scikit-learn, MLflow (local tracking).

4. Action Recommendation Engine

- Map predicted risk categories to personalized interventions.

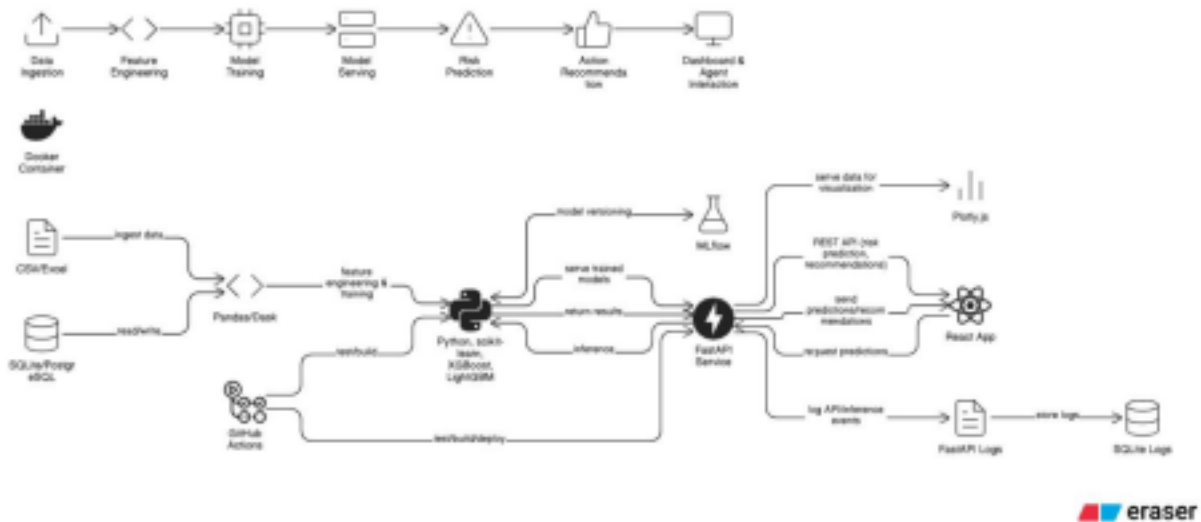
5. Serving Layer & API

- Expose model predictions via FastAPI service.
- Containerization with Docker for portability.

6. Visualization & Dashboard

- Web UI for portfolio .
- Tools: React.js frontend, Plotly.js for interactive visualizations.

Architecture Diagram



Tech Stack

- **Data Storage & Processing:** CSV/Excel, SQLite/PostgreSQL, Pandas, Dask
- **ML & MLOps:** Python, scikit-learn, XGBoost, LightGBM, MLflow (local)
- **API & Services:** FastAPI (REST service), Docker
- **Frontend:** React.js + Plotly.js for charts
- **Monitoring & Logging:** Built-in FastAPI logging + SQLite logs
- **CI/CD:** GitHub Actions (optional for testing & builds)

Milestones

Milestone 1: Requirements & Data Collection

Objective: Finalize project requirements and prepare sample datasets.

- **Deliverables:**
 - Documented KPIs.
 - GitHub repository setup.
 - Sample datasets collected.
- **Evaluation:**
 - Data arranged in required fields from datasets.

Milestone 2: Data Ingestion & Initial EDA

Objective: Build ingestion pipeline and perform exploratory analysis.

- **Deliverables:**

- CSV to SQLite/PostgreSQL ingestion pipeline.
- EDA report.
- **Evaluation:**
 - Successful ingestion verified.
 - EDA report submitted.

Milestone 3: Baseline Model Development

Objective: Train initial Logistic Regression model.

- **Deliverables:**
 - Feature engineering pipeline.
 - Logistic Regression baseline model.
- **Evaluation:**
 - Baseline AUC-ROC measured.
 - If accuracy is low, refine feature pipeline.

Milestone 4: Advanced Model Training

Objective: Develop high-performance ML models.

- **Deliverables:**
 - XGBoost and LightGBM models.
 - Hyperparameter tuning results.
- **Evaluation:**
 - Improved AUC-ROC compared to baseline.

Milestone 5: Recommendation Engine & API Prototype

Objective: Deliver actionable borrower recommendations.

- **Deliverables:**
 - Action recommendation engine.
 - FastAPI prototype for risk scoring.
- **Evaluation:**
 - Local API tested successfully.
 - Recommendations validated for logic.

Milestone 6: Frontend Development & Final Delivery

Objective: Build dashboards and validate system with stakeholders.

- **Deliverables:**
 - React.js frontend with Plotly.js charts.
 - Agent recommendation dashboards.

- Documentation and rollout plan.
- **Evaluation:**
 - UAT confirms test queries display correct recommendations.
 - Project deemed successful